

JINGXUAN WANG

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Personal Information

Chinese Name: 王景煊 **Major:** Software Engineering **English:** IELTS 7.0

Research Interests

AI for Drug Discovery | Precision Medicine | Multimodal Learning in Healthcare | Single-cell & Omics Data Analysis | Spatial Transcriptomics

Education

Beijing University of Posts and Telecommunications (BUPT)

Beijing, China

B.Eng. in Software Engineering

Sep 2023 – Jun 2027

GPA: 88.61 / 100 3.59 / 4.00

Core Courses: Data Structures & Algorithms, Machine Learning, Deep Learning, Artificial Intelligence, Data Mining, Mathematical Statistics, Optimization Methods, Database Systems, Software Engineering

Publications

- Wang, J.**, Yang, J., Khan, M. A., Yee, P. L., Baili, J., Hu, D. DGAN-MPCC: A Novel Dual-GAN Enhanced Multi-Positive Contrastive Clustering Method for Omics Data. *IEEE Journal of Biomedical and Health Informatics*, 2025. (Accepted, JCR Q1). DOI: 10.1109/JBHI.2025.3631973
- Wang, J.**, Guan, R., Liang, K., Wang, S., Yang, X., Cui, X., Hu, D. BiFlow-GAN: Integrating Bidirectional Flows and Cross-Modal Enhancement for Multi-Modal Spatial Transcriptomics. *IEEE Transactions on Multimedia* (Under Review, Submitted Apr 2026, JCR Q1).
- Zhu, S., **Wang, J.**, Hu, X., Tao, J. Enhancing Study-Abroad English Learning with AI-Powered Scenario Simulation. In Proceedings of ICFULL 2025 & GLoCALL 2025, 2025. (Oral/Poster Presentation)

Patents & Software Copyright

- Wang, J.**, Tao, X. ListenSpeakStar: A Multi-Agent Collaborative Intelligent Training System for English Listening and Speaking. Software Copyright Registration, China Copyright Protection Center, 2025. Reg. No.: 2025SR2109214.

Research Experience

Multi-Modal Spatial Transcriptomics Analysis

Nov 2025 – May 2026

- Proposed BiFlow-GAN, a bidirectional-flow generative adversarial framework for robust tissue domain identification in spatial transcriptomics.
- Designed a bidirectional cross-modal attention flow (Bi-AF) mechanism with reciprocal Image-to-Gene (I2G) and Gene-to-Image (G2I) pathways for dynamic feature refinement.
- Benchmarked on diverse ST datasets, significantly outperforming state-of-the-art methods in spatially coherent domain identification and biological interpretability.
- Manuscript submitted to IEEE Transactions on Multimedia in April 2026. (Under Review).

AI-driven Omics Data Clustering & Representation Learning

Dec 2024 – Nov 2025

- Proposed a dual-GAN framework to enhance both input and latent representations for noisy single-cell data.
- Designed a multi-positive contrastive clustering strategy to model continuous cell-state transitions.
- Evaluated on multiple real-world single-cell datasets, achieving state-of-the-art clustering performance.
- Published in IEEE Journal of Biomedical and Health Informatics (JCR Q1, 2025).

AI-Powered Scenario Simulation for Language Learning

Apr 2025 – Jul 2025

- Conducted an undergraduate innovation research project focused on AI-driven scenario simulation for study-abroad English learning.

- Applied statistical analysis and visualization to evaluate learning outcomes across different simulated scenarios, identifying patterns in learner performance and engagement.
- Resulted in a peer-reviewed conference paper and presentation at ICFULL 2025 & GLoCALL 2025.

ListenSpeakStar: Multi-Agent Collaborative English Training System

May 2025 – Jan 2026

- Developed a multi-agent workflow for English listening and speaking training, integrating scenario generation, learner interaction, error diagnosis, adaptive feedback, and performance evaluation agents. Built a modular orchestration framework to coordinate multi-turn interactions among agents, enabling personalized practice sessions and structured learning feedback.
- Achieved software copyright registration for ListenSpeakStar V1.0. Reg. No.: 2025SR2109214.

International Programs & Academic Exchange

SOC Summer Workshop 2025 — National University of Singapore (NUS)

May – Jul 2025

- Course: Big Data Analytics and Visualization. Grade: A+. Programme intensity equivalent to a regular semester course at NUS School of Computing.

OxCam Programme 2024 — University of Oxford & University of Cambridge

Aug 2024

- AI + Biotechnology Engineering & Healthcare Technology Track. Overall Evaluation Score: A-. Focused on AI-driven healthcare technology and biomedical applications.

Beijing Medical Innovation Center Winter Training Program 2026

Feb 2026

- Selected as participant; focused on medical innovation and translational research.
- Delivered comprehensive presentation on Alzheimer's Disease covering pathological mechanisms.
- Awarded “Outstanding Trainee” for exceptional performance in group presentations and academic discussions.

Competitions & Awards

- China Students Service Outsourcing Innovation & Entrepreneurship Competition (15th Ed.) — **National Third Prize**, 2024. Cert. No.: F2403243A3293.
- Global Campus AI Algorithm Elite Competition (6th Ed.), Beijing Regional Final, AI + Medicine Track — **Second Prize**, 2024. Project: Medical Ollama.
- China Students Service Outsourcing Innovation & Entrepreneurship Competition (16th Ed.), Beijing Regional — **Provincial Third Prize**, 2025. Cert. No.: NT2402245A3068.
- National Undergraduate Statistical Modeling Competition (11th Ed.), Beijing — **Regional First Prize**, 2025. Project: Spatiotemporal Efficiency Evaluation and Multi-objective Simulation for Medical Resource Allocation. Cert. No.: 20251101A0080.
- China Undergraduate Computer Design Competition (18th Ed.), Beijing Municipal — **Regional Second Prize**, 2025. Project: Zhang Heng Mathematics AI-Agent. Project ID: 2025003391.
- National Artificial Intelligence Application Innovation Competition, University-Level — **Excellence Award**, 2025. Team: Fat Orange Thinking.
- University Third-class Scholarship.

Skills

Programming & Development: Python, PyTorch, R, SQL, Git, Linux

Machine Learning & AI: Deep Learning, Contrastive Learning, Generative Models (GANs), Graph Neural Networks (GNN), Vision Transformer (ViT), Representation Learning, RAG, LLMs, Multi-Agent Systems

Biomedical & Data Science: Single-cell & Omics Data Analysis, Spatial Transcriptomics (10x Visium, Stereo-seq), Multimodal Learning in Healthcare, Biomedical Text Mining

Mathematics & Modeling: Mathematical Statistics, Optimization Methods, Statistical Modeling

Tools & Research Utilities: LaTeX, Jupyter Notebook, Data Visualization